DEVOPS CAPSTONE

GitTogether

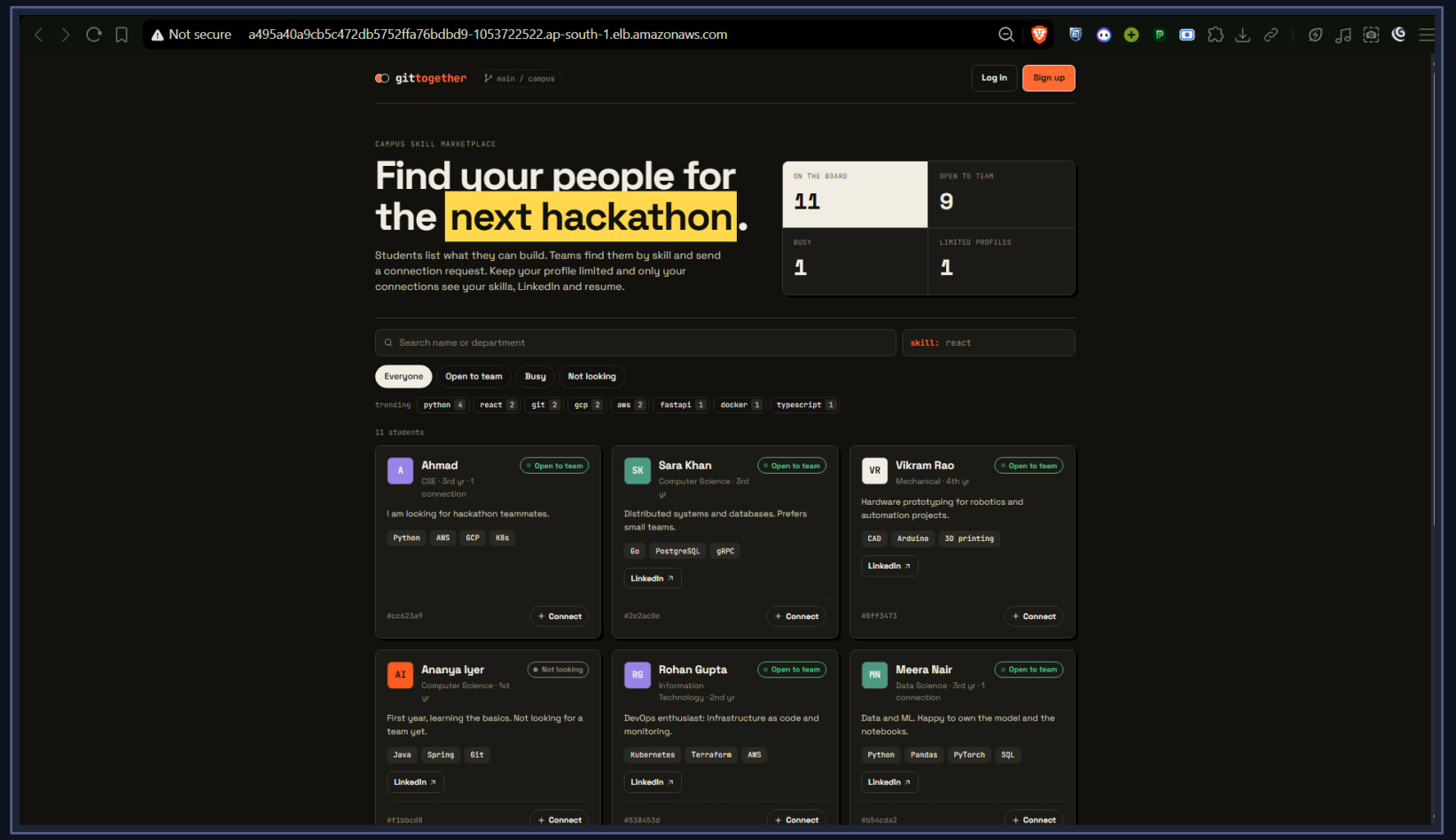
A teammate-finder for students, taken from a laptop to a monitored Kubernetes cluster on AWS

Ahmad Anzar

github.com/AhmadAnzar/GitTogether

THE APP

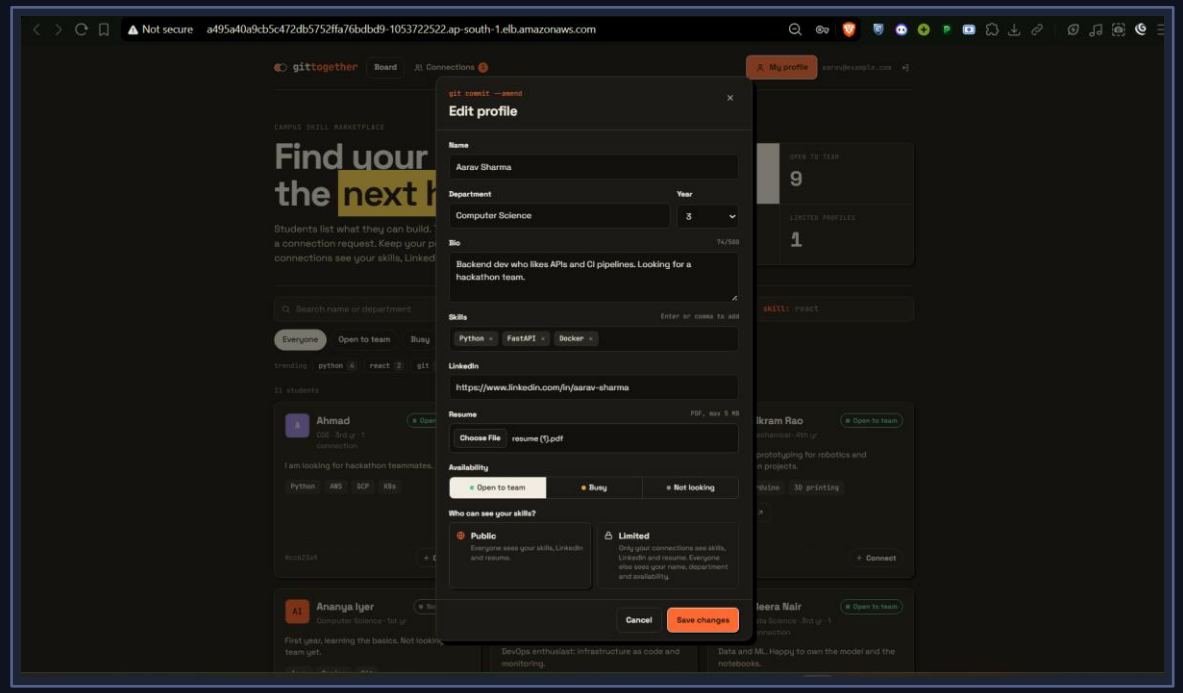
Find teammates for projects and hackathons

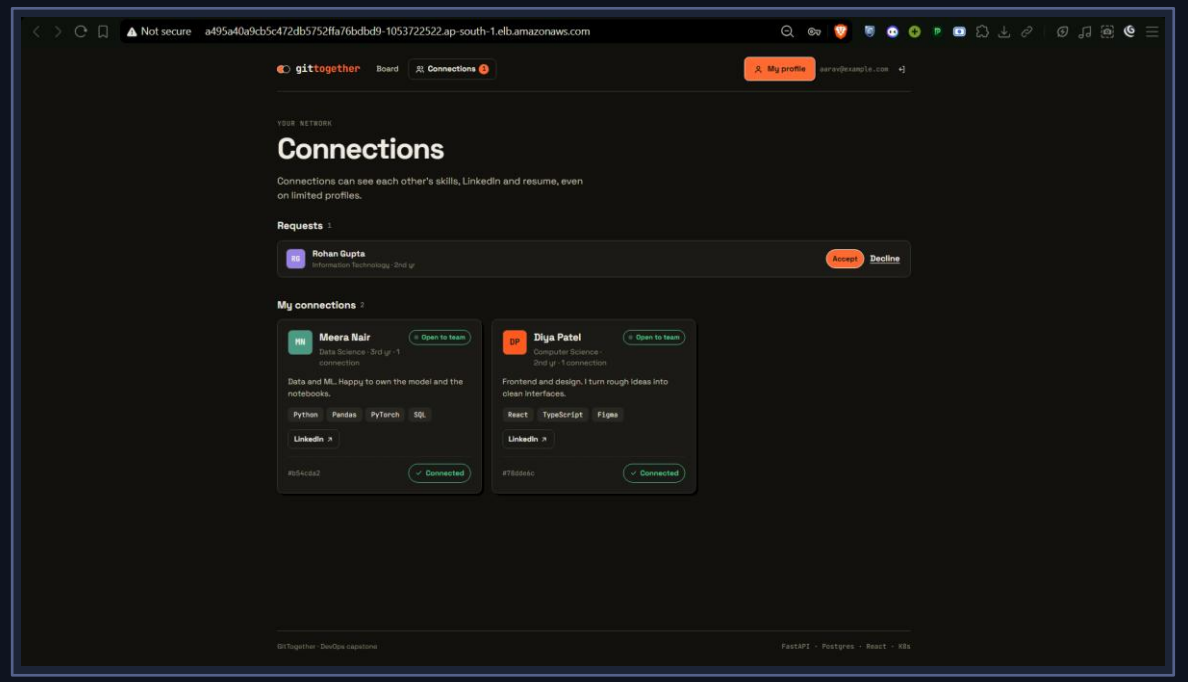


Make a profile with skills and availability, browse students, **send connection requests**.

THE APP

Profiles and connections

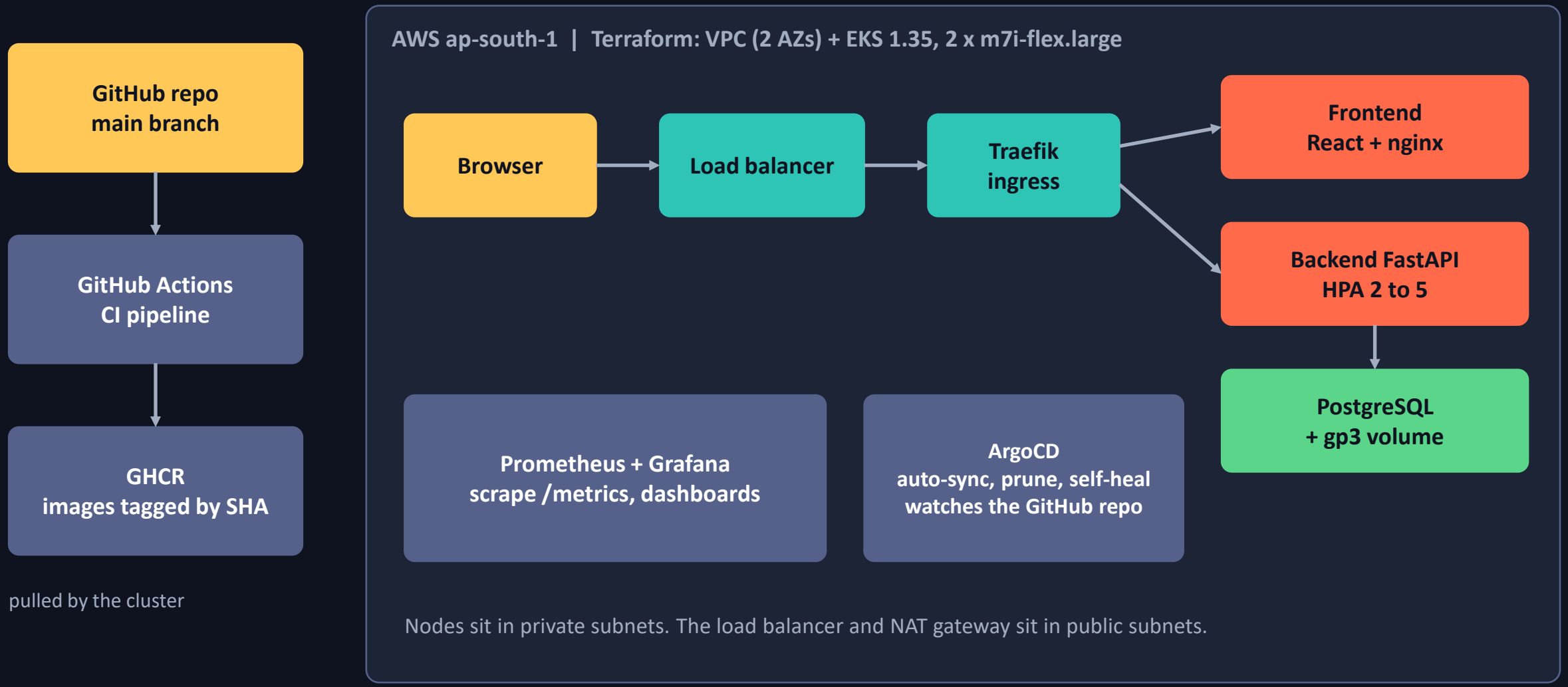




Public profiles with skills and availability. **Connection requests** can be sent, accepted and tracked.

ARCHITECTURE

From git push to a running cluster



TECH STACK

What it is built with

APP

React + Vite, FastAPI, SQLAlchemy, PostgreSQL

QUALITY

pytest with a 90% coverage gate, ruff, ESLint, Vitest

SECURITY

gitleaks, Trivy, Dependabot, CodeQL

CONTAINERS

Multi-stage Docker builds, non-root users

CI/CD

GitHub Actions, GHCR, SHA-tagged images

INFRASTRUCTURE

Terraform: VPC, EKS, KMS, IAM, Pod Identity

KUBERNETES

Helm, Traefik, HPA, Pod Security (restricted)

OPERATIONS

ArgoCD, Prometheus, Grafana

TESTING

65 tests pass, 99% coverage

```
============================================================ warnings summary ============================================================
..\..\..\..\..\miniconda3\Lib\site-packages\fastapi\testclient.py:1
  C:\Users\mdanz\miniconda3\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using `httpx` with `starlette.testclient` is deprecated;
 install `httpx2` instead.
    from starlette.testclient import TestClient as TestClient  # noqa

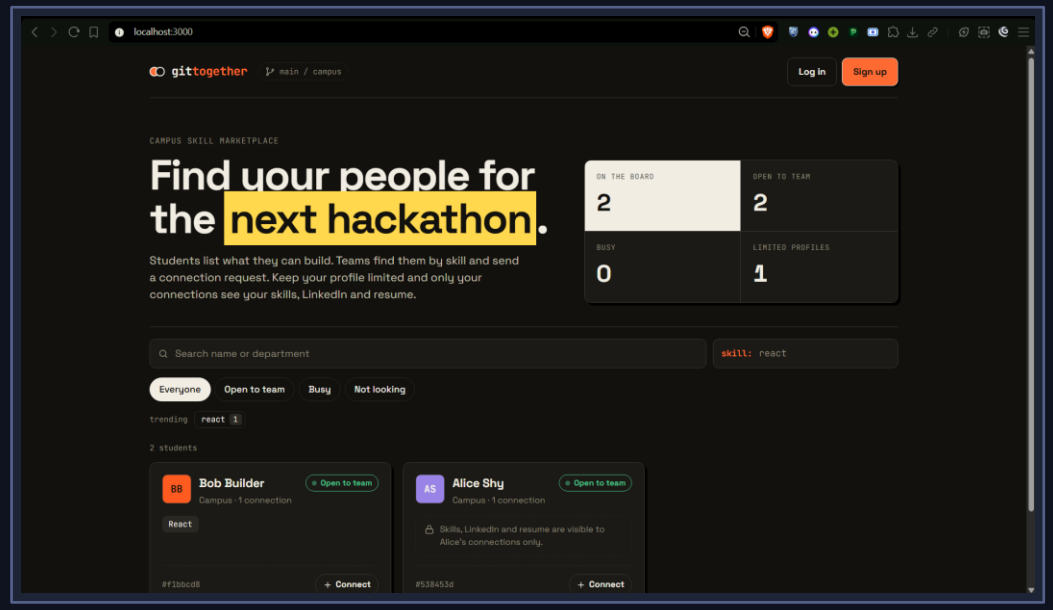
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
============================================================ tests coverage ============================================================
_____________________________________________ coverage: platform win32, python 3.13.13-final-0 _____________________________________________

Name                          Stmts   Miss  Cover   Missing
-----------------------------------------------------------
app\__init__.py                   0      0   100%
app\access.py                    50      0   100%
app\config.py                    10      0   100%
app\db.py                        12      0   100%
app\main.py                      27      0   100%
app\models.py                    41      0   100%
app\routers\__init__.py           0      0   100%
app\routers\auth.py              39      0   100%
app\routers\connections.py       87      0   100%
app\routers\profiles.py         127      1    99%   139
app\schemas.py                  114      2    98%   25, 39
app\security.py                  36      0   100%
app\serializers.py               18      0   100%
-----------------------------------------------------------
TOTAL                           561      3    99%
Required test coverage of 90% reached. Total coverage: 99.47%
============================================================ 65 passed, 1 warning in 9.10s =============================================================
```

65 passed, 99.47% coverage against a **90%** gate, on a separate test database.

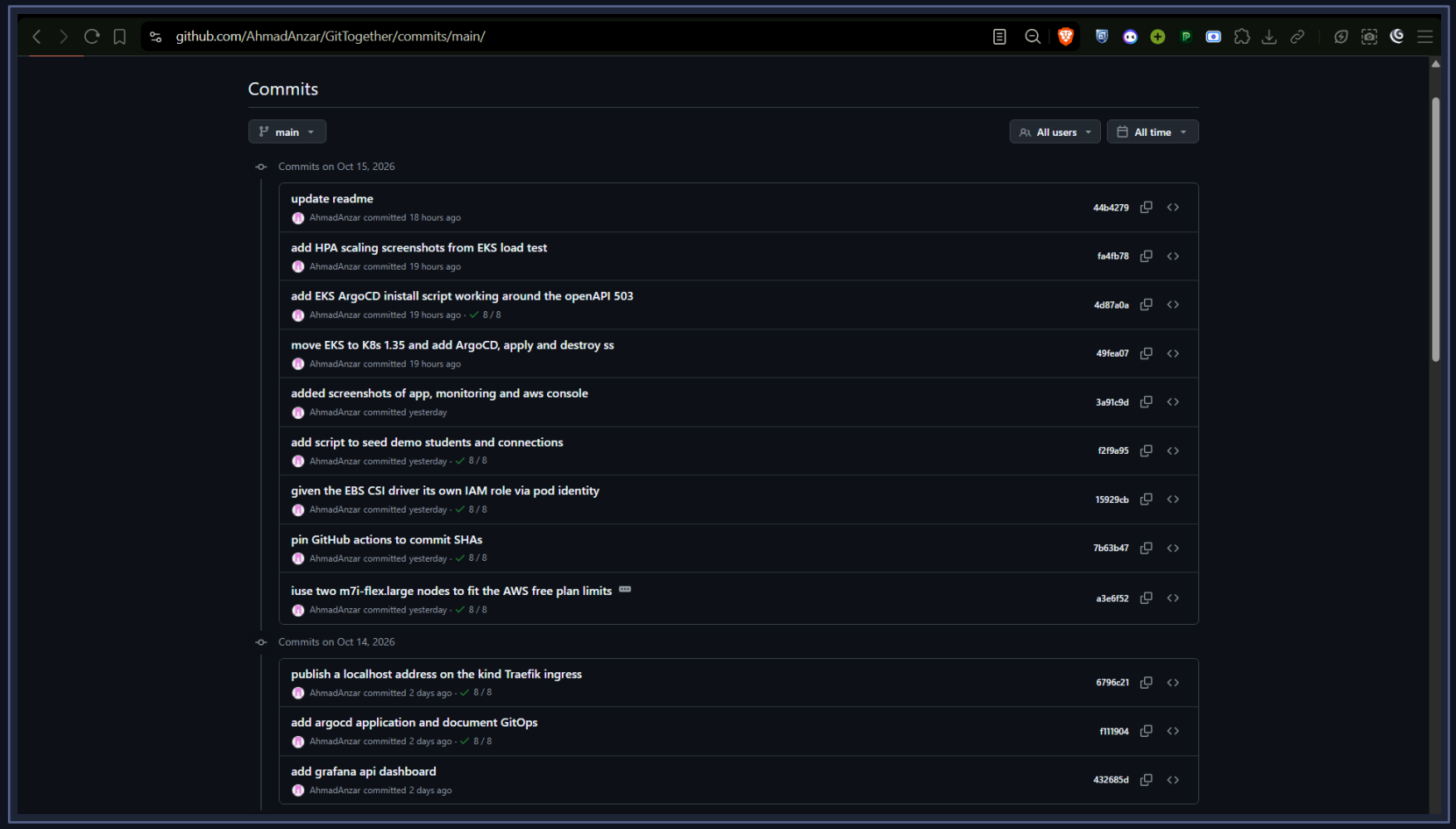
One command starts the whole stack

```
(base) PS C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>  docker compose ps
NAME                      IMAGE                     COMMAND                  SERVICE    CREATED          STATUS                    PORTS
gittogether-backend-1     gittogether-backend       "sh -c 'alembic upgr…"   backend    4 minutes ago    Up 4 minutes (healthy)    127.0.0.1:8000->8000/tcp
gittogether-frontend-1    gittogether-frontend      "/docker-entrypoint.…"   frontend   4 minutes ago    Up 4 minutes (healthy)    0.0.0.0:3000->8080/tcp, [::]:30
00->8080/tcp
gittogether-postgres-1    postgres:16-alpine        "docker-entrypoint.s…"   postgres   4 minutes ago    Up 4 minutes (healthy)    5432/tcp
(base) PS C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>
```



docker compose up --build: frontend, backend and Postgres, all healthy.

48 commits on a public repo

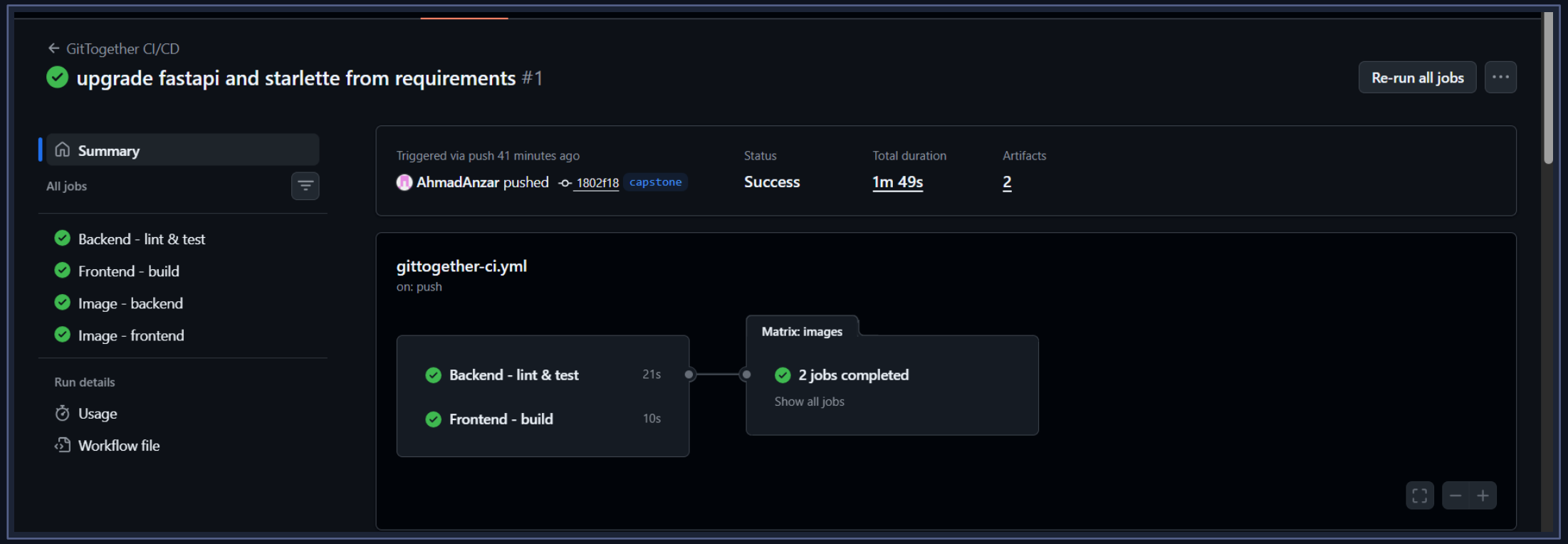


Specific commit messages on the main branch, with a green check on each pipeline run.

CI/CD

Every push is tested, scanned and published

Secret scan → Lint + tests → Terraform + Helm → Docker build → Trivy gate → Push to GHCR



65 backend tests, **90%** coverage gate, **7** jobs, images tagged by commit SHA

The run summary

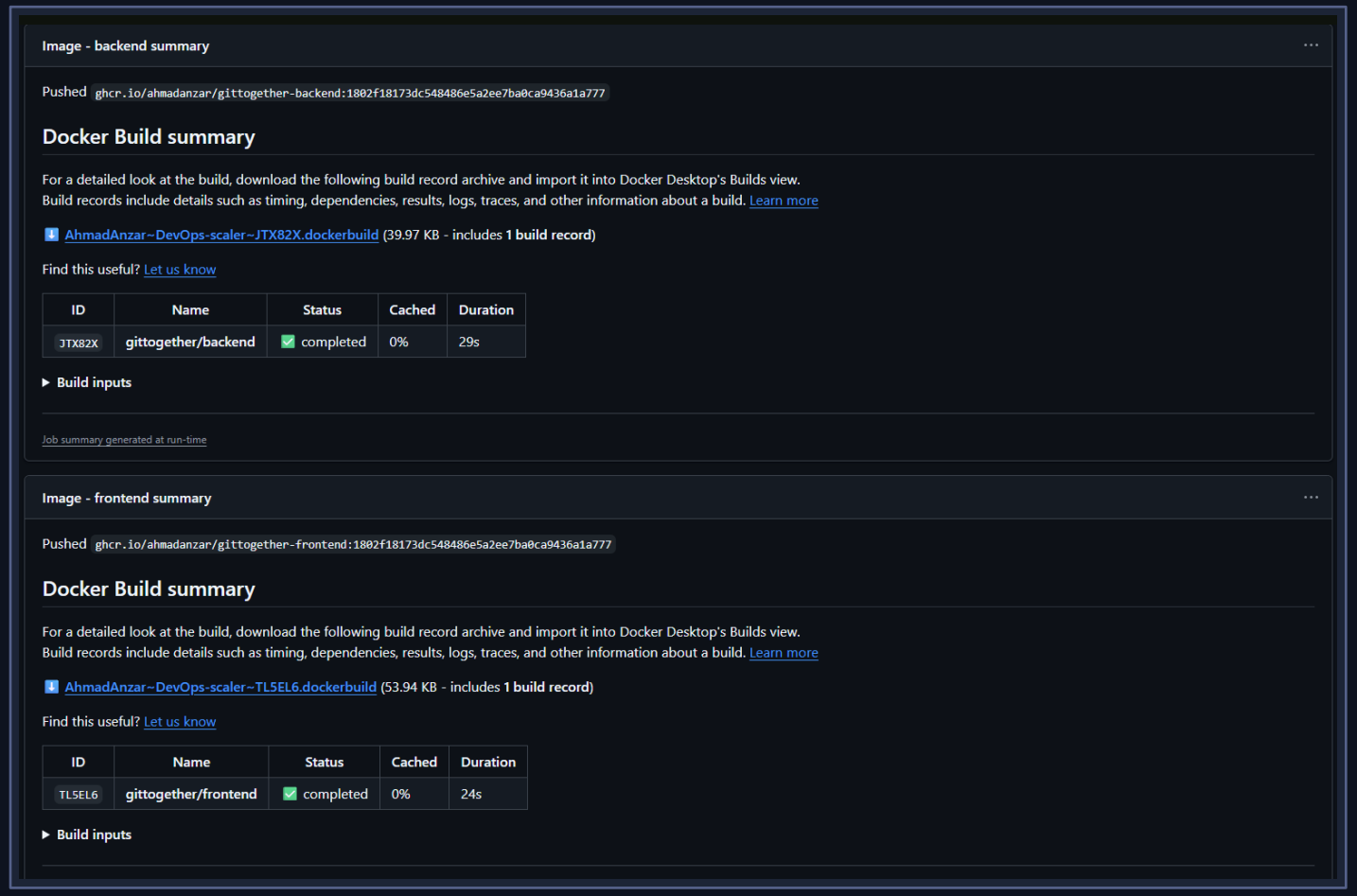


Image - backend summary

Pushed `ghcr.io/ahmadanzar/gittogether-backend:1802f18173dc548486e5a2ee7ba0ca9436a1a777`

Docker Build summary

For a detailed look at the build, download the following build record archive and import it into Docker Desktop's Builds view.
Build records include details such as timing, dependencies, results, logs, traces, and other information about a build. Learn more

AhmadAnzar~DevOps-scaler~JTX82X.dockerbuild (39.97 KB - includes **1 build record**)

Find this useful? Let us know

ID	Name	Status	Cached	Duration
JTX82X	gittogether/backend	completed	0%	29s

▶ Build inputs

Job summary generated at run-time

Image - frontend summary

Pushed `ghcr.io/ahmadanzar/gittogether-frontend:1802f18173dc548486e5a2ee7ba0ca9436a1a777`

Docker Build summary

For a detailed look at the build, download the following build record archive and import it into Docker Desktop's Builds view.
Build records include details such as timing, dependencies, results, logs, traces, and other information about a build. Learn more

AhmadAnzar~DevOps-scaler~TL5EL6.dockerbuild (53.94 KB - includes **1 build record**)

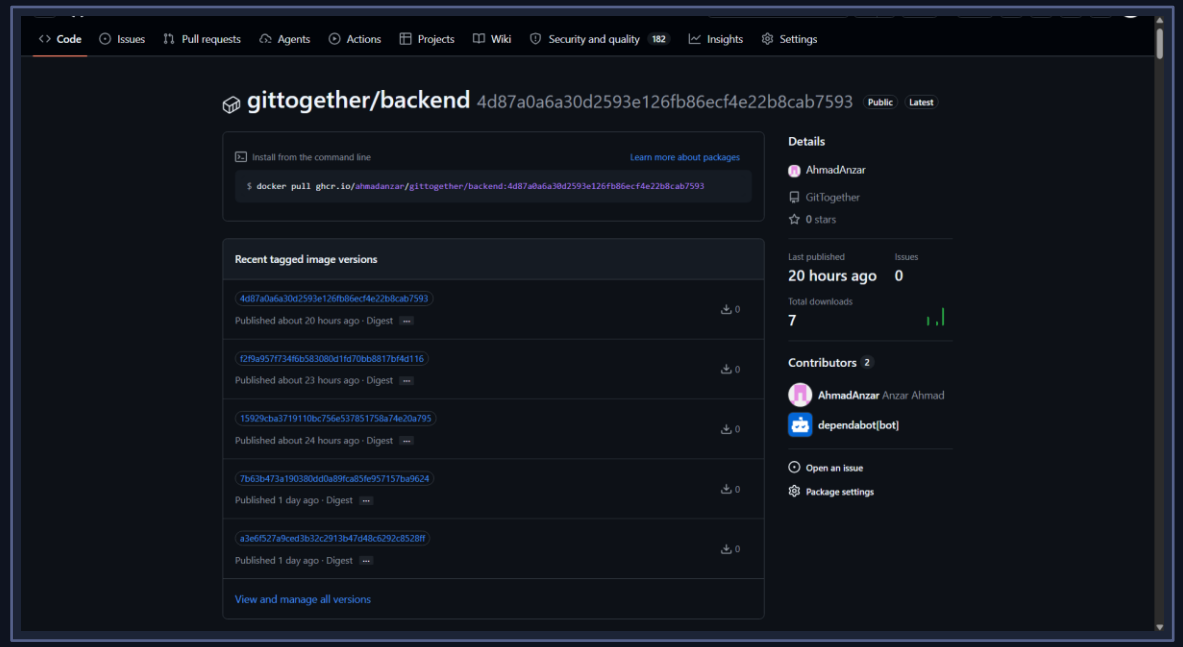
Find this useful? Let us know

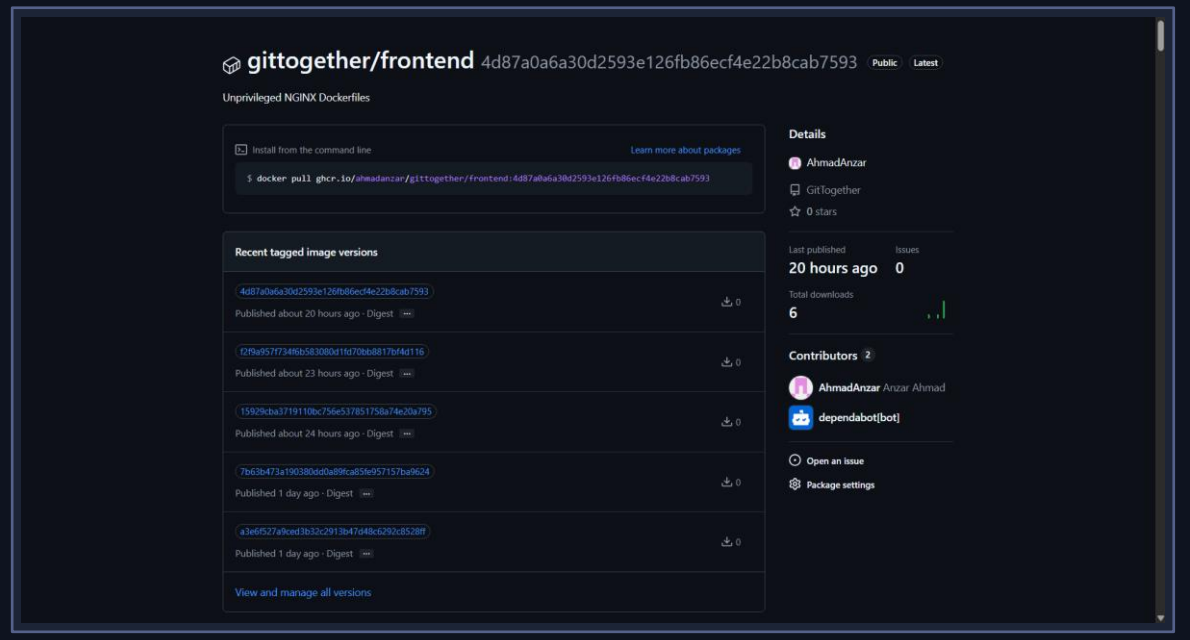
ID	Name	Status	Cached	Duration
TL5EL6	gittogether/frontend	completed	0%	24s

▶ Build inputs

Backend and frontend jobs **in parallel**, then one image job per component.

Images in the GitHub Container Registry

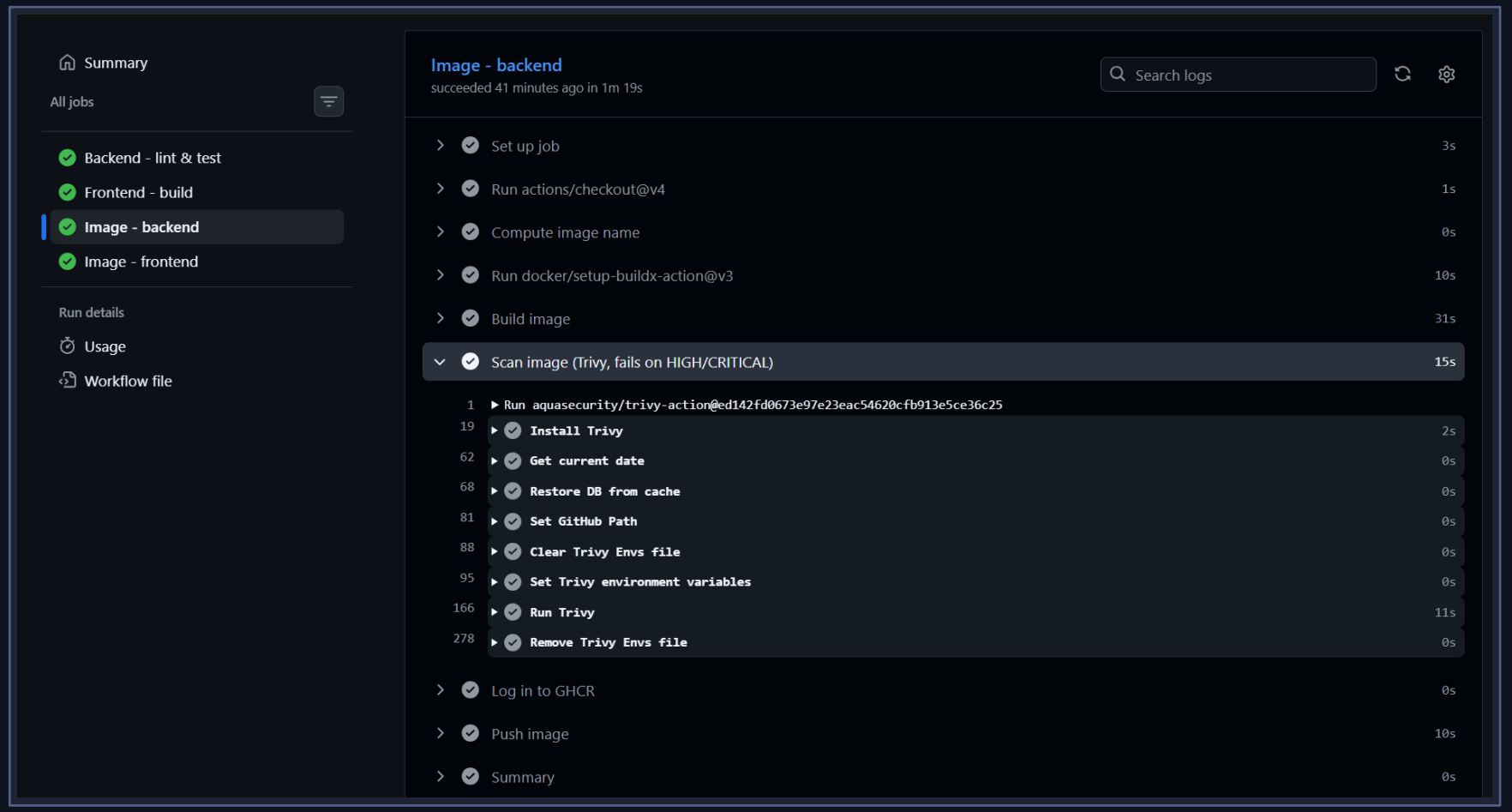




Tagged with the commit SHA, never latest, so every deployment traces back to a commit.

SECURITY

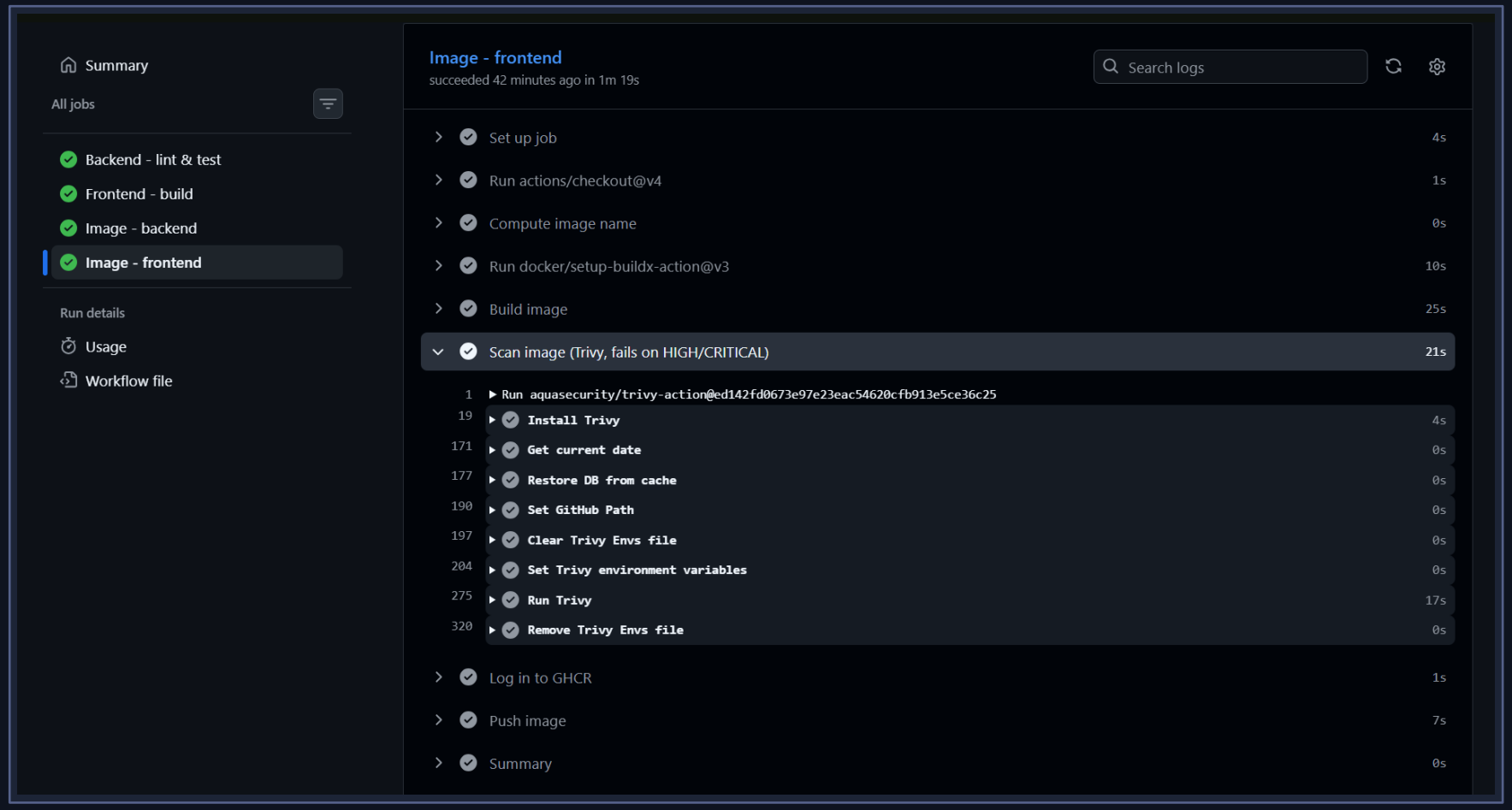
Trivy on the backend image



Fails the build on HIGH/CRITICAL. The first run found 3 HIGH CVEs in Starlette; upgrading fixed them.

SECURITY

Trivy on the frontend image



41 issues in the old nginx base image, **zero** after moving to nginx-unprivileged 1.30.

INFRASTRUCTURE

One terraform apply builds AWS

```
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]: Still creating... [00m10s elapsed]
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]: Still creating... [00m20s elapsed]
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]: Still creating... [00m30s elapsed]
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]: Still creating... [00m40s elapsed]
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]: Still creating... [00m50s elapsed]
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]: Still creating... [01m00s elapsed]
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]: Still creating... [01m10s elapsed]
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0]: Creation complete after 1m19s [id=gittogether-eks:main-38741b0463ceb06ff696e7d316]
module.eks.aws_eks_addon.this["eks-pod-identity-agent"]: Creating...
module.eks.aws_eks_addon.this["coredns"]: Creating...
module.eks.aws_eks_addon.this["metrics-server"]: Creating...
module.eks.aws_eks_addon.this["kube-proxy"]: Creating...
module.eks.aws_eks_addon.this["aws-ebs-csi-driver"]: Creating...
module.eks.aws_eks_addon.this["coredns"]: Still creating... [00m10s elapsed]
module.eks.aws_eks_addon.this["metrics-server"]: Still creating... [00m10s elapsed]
module.eks.aws_eks_addon.this["kube-proxy"]: Still creating... [00m10s elapsed]
module.eks.aws_eks_addon.this["eks-pod-identity-agent"]: Still creating... [00m10s elapsed]
module.eks.aws_eks_addon.this["aws-ebs-csi-driver"]: Still creating... [00m10s elapsed]
module.eks.aws_eks_addon.this["coredns"]: Creation complete after 14s [id=gittogether-eks:coredns]
module.eks.aws_eks_addon.this["metrics-server"]: Still creating... [00m20s elapsed]
module.eks.aws_eks_addon.this["kube-proxy"]: Still creating... [00m20s elapsed]
module.eks.aws_eks_addon.this["aws-ebs-csi-driver"]: Still creating... [00m20s elapsed]
module.eks.aws_eks_addon.this["eks-pod-identity-agent"]: Still creating... [00m20s elapsed]
module.eks.aws_eks_addon.this["eks-pod-identity-agent"]: Creation complete after 24s [id=gittogether-eks:eks-pod-identity-agent]
module.eks.aws_eks_addon.this["kube-proxy"]: Creation complete after 24s [id=gittogether-eks:kube-proxy]
module.eks.aws_eks_addon.this["metrics-server"]: Still creating... [00m30s elapsed]
module.eks.aws_eks_addon.this["aws-ebs-csi-driver"]: Still creating... [00m30s elapsed]
module.eks.aws_eks_addon.this["aws-ebs-csi-driver"]: Creation complete after 36s [id=gittogether-eks:aws-ebs-csi-driver]
module.eks.aws_eks_addon.this["metrics-server"]: Still creating... [00m40s elapsed]
module.eks.aws_eks_addon.this["metrics-server"]: Creation complete after 45s [id=gittogether-eks:metrics-server]

Apply complete! Resources: 59 added, 0 changed, 0 destroyed.

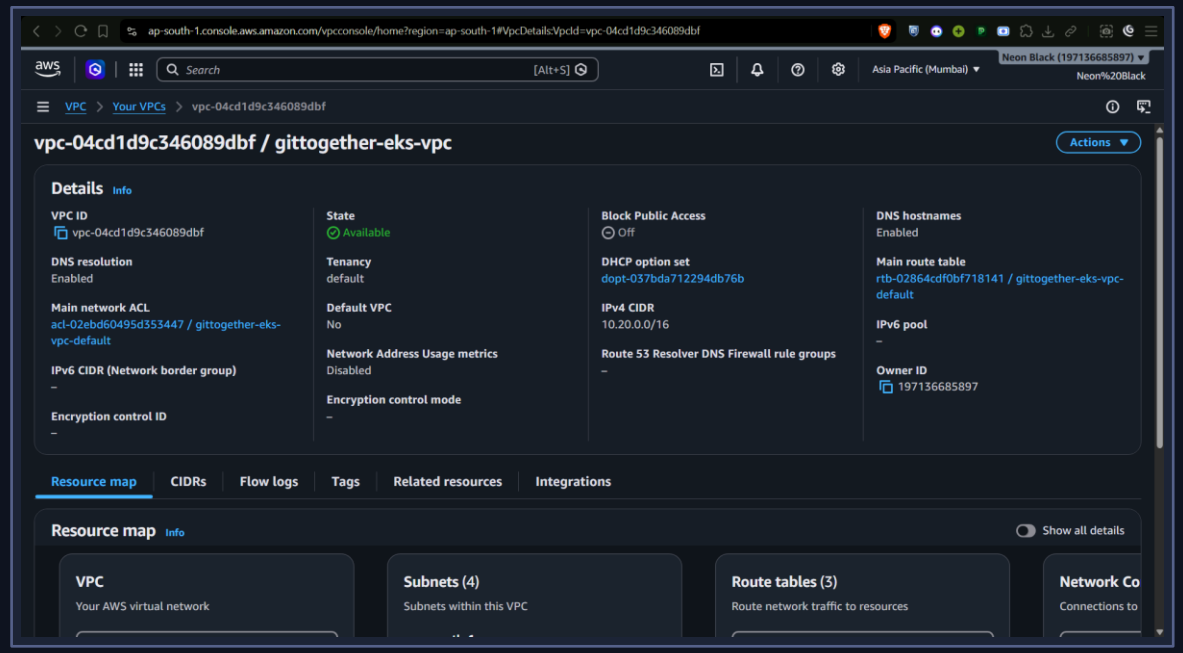
Outputs:

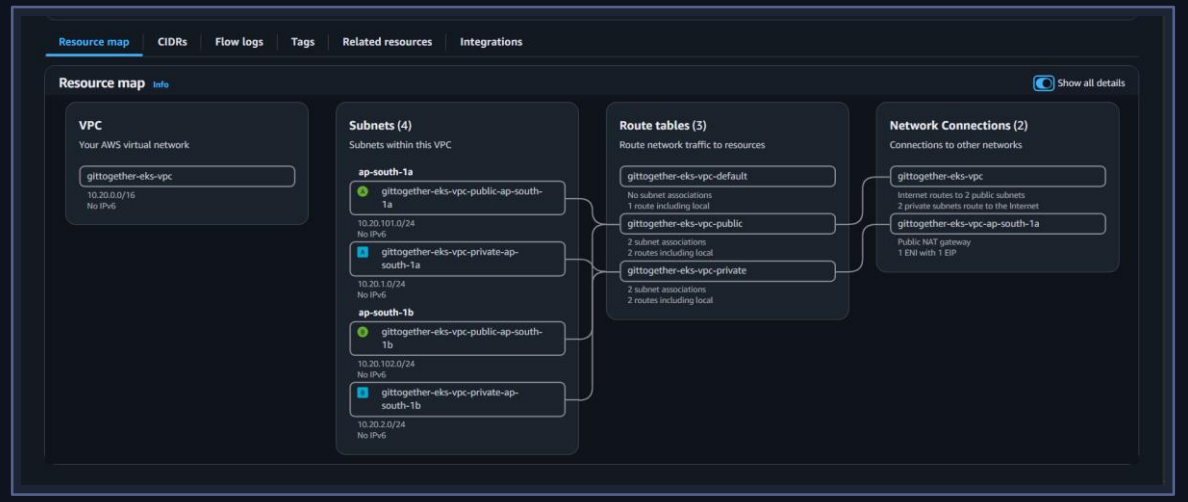
cluster_endpoint = "https://9294070E6388CE89BD2469A20D29A208.gr7.ap-south-1.eks.amazonaws.com"
cluster_name = "gittogether-eks"
configure_kubectl = "aws eks update-kubeconfig --region ap-south-1 --name gittogether-eks"
kubernetes_version = "1.35"
private_subnet_ids = [
  "subnet-07012eebcd9ce0823",
  "subnet-0821c5828a1b86754",
]
public_subnet_ids = [
  "subnet-05f0d9b567350ca54",
  "subnet-0c8eb1a7b84ec0be2",
]
region = "ap-south-1"
vpc_id = "vpc-0263145a7934ba574"

C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project\terraform>
```

59 resources created in about 15 minutes: network, cluster, nodes, add-ons.

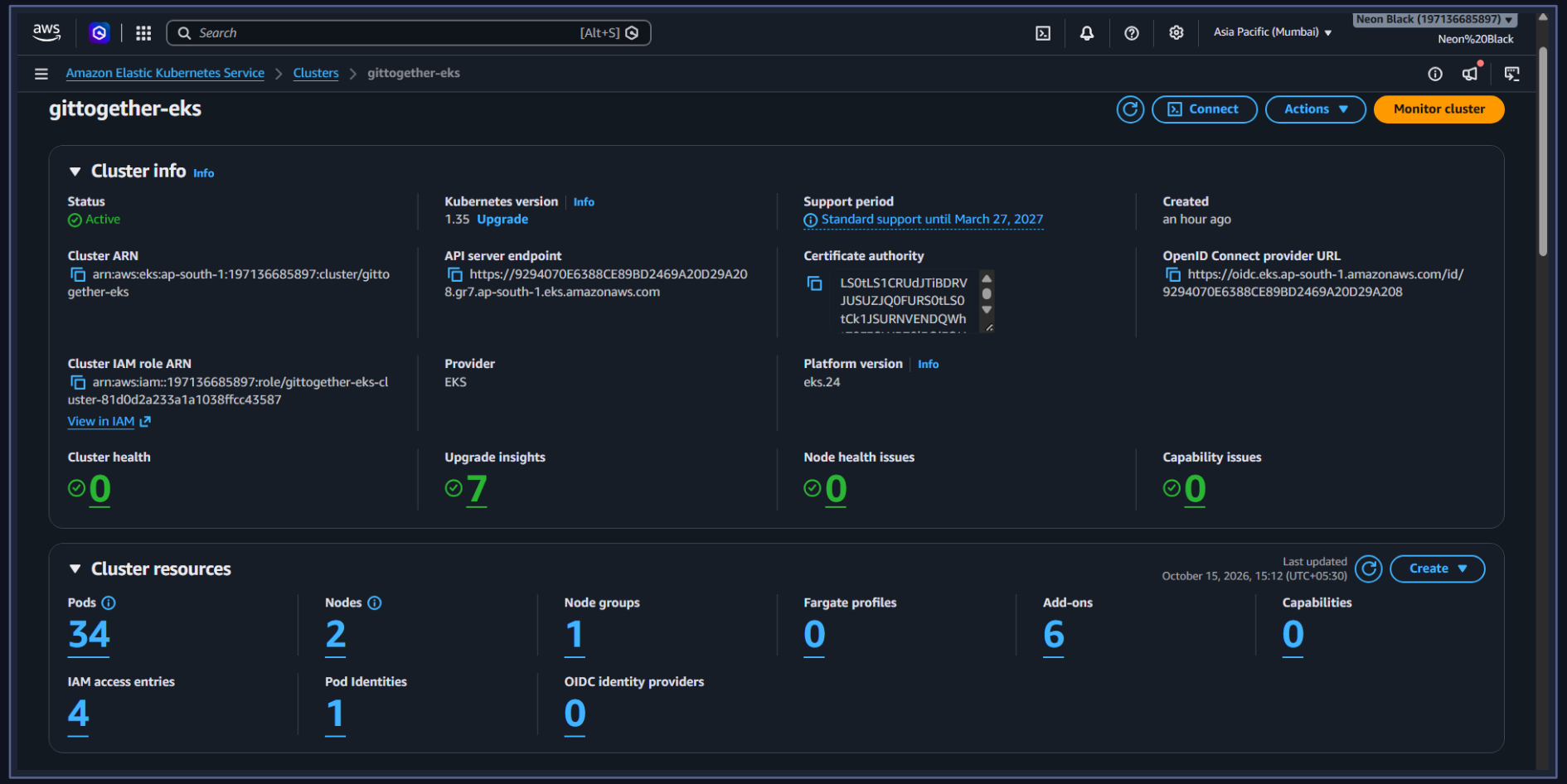
VPC and subnets





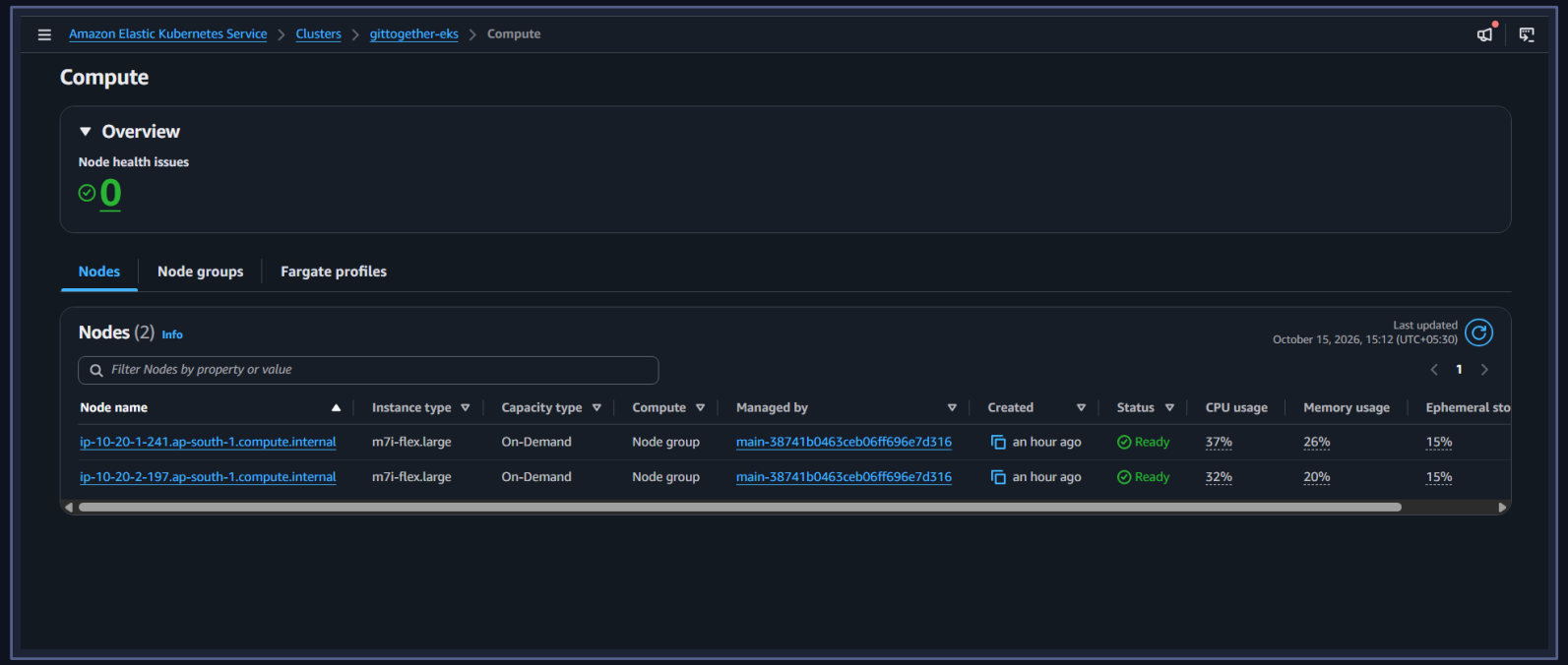
Public subnets for the load balancer and NAT, **private subnets** for the nodes.

The EKS cluster



EKS 1.35, standard support, 0 health issues, 6 add-ons, KMS-encrypted secrets.

Two worker nodes




```
C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project> kubectl get nodes -o wide
NAME                                          STATUS   ROLES    AGE   VERSION               INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                        KERNEL-VERSION
                CONTAINER-RUNTIME
ip-10-20-1-241.ap-south-1.compute.internal    Ready    <none>   48m   v1.35.8-eks-b9510e3   10.20.1.241   <none>        Amazon Linux 2023.12.20261012   6.12.111-136.204.am
zn2023.x86_64   containerd://2.3.5+unknown
ip-10-20-2-197.ap-south-1.compute.internal    Ready    <none>   48m   v1.35.8-eks-b9510e3   10.20.2.197   <none>        Amazon Linux 2023.12.20261012   6.12.111-136.204.am
zn2023.x86_64   containerd://2.3.5+unknown

C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>
```

2 x m7i-flex.large, both Ready, running Kubernetes v1.35.

Everything running

```
C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>kubectl get pods -n gittogether
NAME                                  READY   STATUS    RESTARTS   AGE
gittogether-backend-c57f75f46-ff2gr   1/1     Running   0          32m
gittogether-backend-c57f75f46-szkbq   1/1     Running   0          33m
gittogether-frontend-6d7c445784-6q8bg   1/1     Running   0          33m
gittogether-frontend-6d7c445784-6tlh5   1/1     Running   0          33m
gittogether-postgres-0                1/1     Running   0          33m

C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>kubectl get svc -n gittogether
NAME                  TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
gittogether-backend    ClusterIP   172.20.195.122   <none>        8000/TCP   33m
gittogether-frontend   ClusterIP   172.20.128.251   <none>        80/TCP     33m
gittogether-postgres   ClusterIP   None             <none>        5432/TCP   33m

C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>helm list -n gittogether
NAME            NAMESPACE       REVISION        UPDATED                                 STATUS          CHART                   APP VERSION
gittogether     gittogether     1               2026-10-15 11:48:40.766043 +0530 IST    deployed        gittogether-1.1.0       1.1.0

C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>
```

2 backend, 2 frontend, 1 Postgres, deployed from one Helm chart.

Health and metrics

```
C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>curl http://a495a40a9cb5c472db5752ffa76bdbd9-1053722522
.ap-south-1.elb.amazonaws.com/health
{"status":"UP"}
C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>kubectl port-forward svc/gittogether-backend 8000:8000
-n gittogether
Forwarding from 127.0.0.1:8000 -> 8000
Forwarding from [::1]:8000 -> 8000
```

Command Prompt - kubectl p | Windows PowerShell

```
(base) PS C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project> curl.exe -s http://localhost:8000/metrics | Select-Object -First 30
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 21922.0
python_gc_objects_collected_total{generation="1"} 3856.0
python_gc_objects_collected_total{generation="2"} 849.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 267.0
python_gc_collections_total{generation="1"} 24.0
python_gc_collections_total{generation="2"} 2.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="12",patchlevel="15",version="3.12.15"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 4.1078784e+08
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 9.5281152e+07
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.79204515318e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 6.54
(base) PS C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>
```

/health is the liveness check, **/metrics** feeds Prometheus.

The backend scales under load

```
C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>kubectl get hpa -n gittogether -w
NAME                 REFERENCE                               TARGETS        MINPODS   MAXPODS   REPLICAS   AGE
gittogether-backend   Deployment/gittogether-backend   cpu: 2%/70%    2         5         2          53m
gittogether-backend   Deployment/gittogether-backend   cpu: 145%/70%   2         5         2          54m
gittogether-backend   Deployment/gittogether-backend   cpu: 153%/70%   2         5         4          54m
gittogether-backend   Deployment/gittogether-backend   cpu: 141%/70%   2         5         5          55m
gittogether-backend   Deployment/gittogether-backend   cpu: 75%/70%    2         5         5          55m
gittogether-backend   Deployment/gittogether-backend   cpu: 70%/70%    2         5         5          55m
gittogether-backend   Deployment/gittogether-backend   cpu: 67%/70%    2         5         5          55m
gittogether-backend   Deployment/gittogether-backend   cpu: 65%/70%    2         5         5          56m
gittogether-backend   Deployment/gittogether-backend   cpu: 63%/70%    2         5         5          56m
gittogether-backend   Deployment/gittogether-backend   cpu: 55%/70%    2         5         5          56m
gittogether-backend   Deployment/gittogether-backend   cpu: 66%/70%    2         5         5          56m
gittogether-backend   Deployment/gittogether-backend   cpu: 64%/70%    2         5         5          57m
gittogether-backend   Deployment/gittogether-backend   cpu: 70%/70%    2         5         5          57m
gittogether-backend   Deployment/gittogether-backend   cpu: 57%/70%    2         5         5          57m
gittogether-backend   Deployment/gittogether-backend   cpu: 66%/70%    2         5         5          57m
gittogether-backend   Deployment/gittogether-backend   cpu: 65%/70%    2         5         5          58m
gittogether-backend   Deployment/gittogether-backend   cpu: 58%/70%    2         5         5          58m
gittogether-backend   Deployment/gittogether-backend   cpu: 13%/70%    2         5         5          58m
gittogether-backend   Deployment/gittogether-backend   cpu: 2%/70%     2         5         5          58m
```

2 to 5 pods when CPU hit **153%** of the 70% target, then back down when the load stopped.

The pods the autoscaler created

```
C:\Users\mdanz\OneDrive\Desktop\devops\capstone-project>kubectl get pods -n gittogether
NAME                                   READY   STATUS    RESTARTS   AGE
gittogether-backend-7b74df54c7-2bqvh   1/1     Running   0          6m8s
gittogether-backend-7b74df54c7-648md   1/1     Running   0          6m23s
gittogether-backend-7b74df54c7-mlsmv   1/1     Running   0          60m
gittogether-backend-7b74df54c7-qk828   1/1     Running   0          60m
gittogether-backend-7b74df54c7-sxx2q   1/1     Running   0          6m23s
gittogether-frontend-675f5b9f8c-fjfmd  1/1     Running   0          60m
gittogether-frontend-675f5b9f8c-vrlqp  1/1     Running   0          60m
gittogether-postgres-0                 1/1     Running   0          60m
```

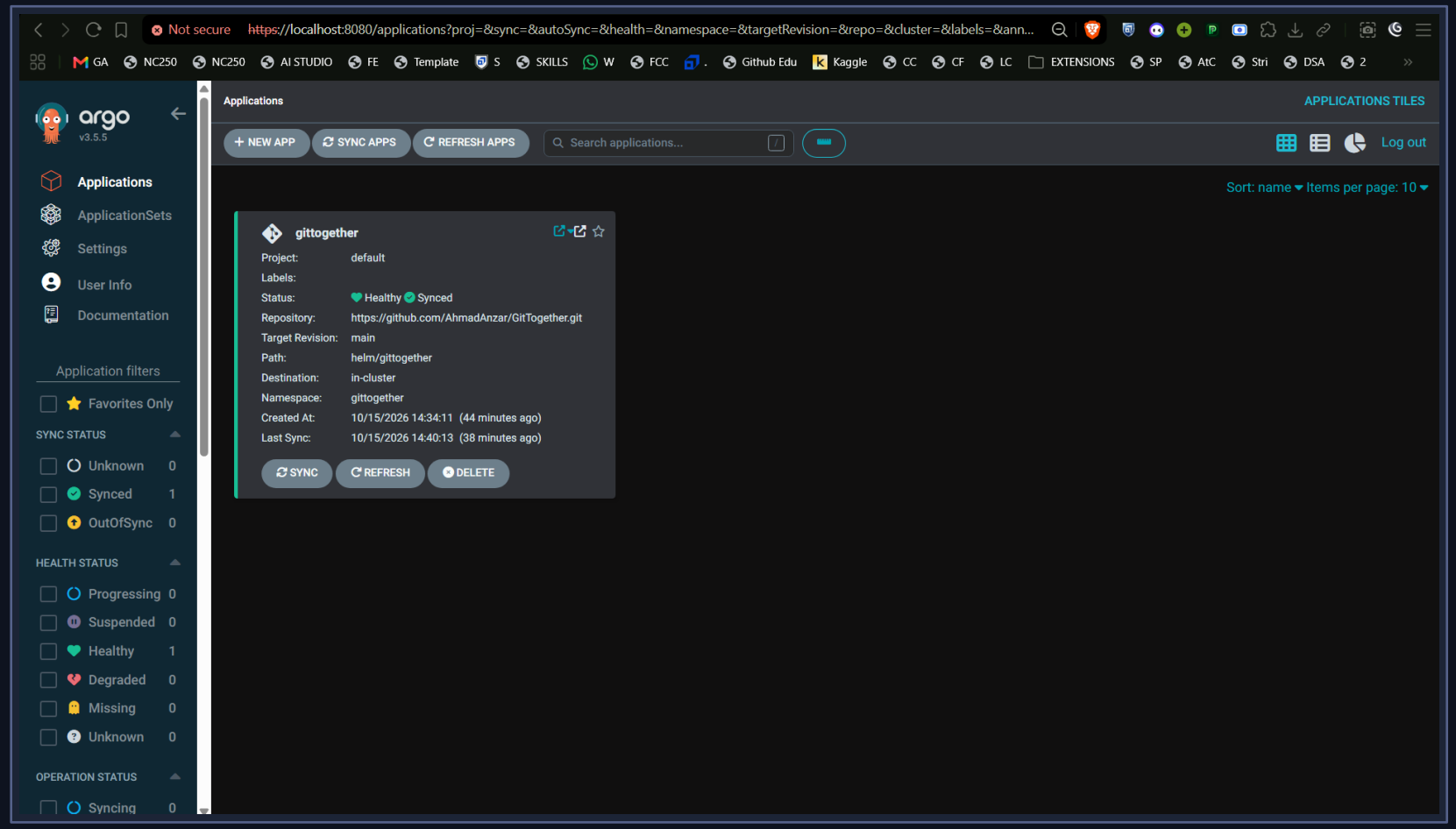
Command Prompt - kubectl p | Windows PowerShell | MINGW64:/c/Users/mdanz/O

```
mdanz@DESKTOP-1CJGKDG MINGW64 ~/OneDrive/Desktop/devops/capstone-project (main)
$ export URL="http://a495a40a9cb5c472db5752ffa76bdbd9-1053722522.ap-south-1.elb.amazonaws.com"
bash scripts/load-test.sh
Hitting http://a495a40a9cb5c472db5752ffa76bdbd9-1053722522.ap-south-1.elb.amazonaws.com/api/profiles?limit=100 with 20 parallel workers for 120s...
Load test completed: 7175 requests.

mdanz@DESKTOP-1CJGKDG MINGW64 ~/OneDrive/Desktop/devops/capstone-project (main)
$
```

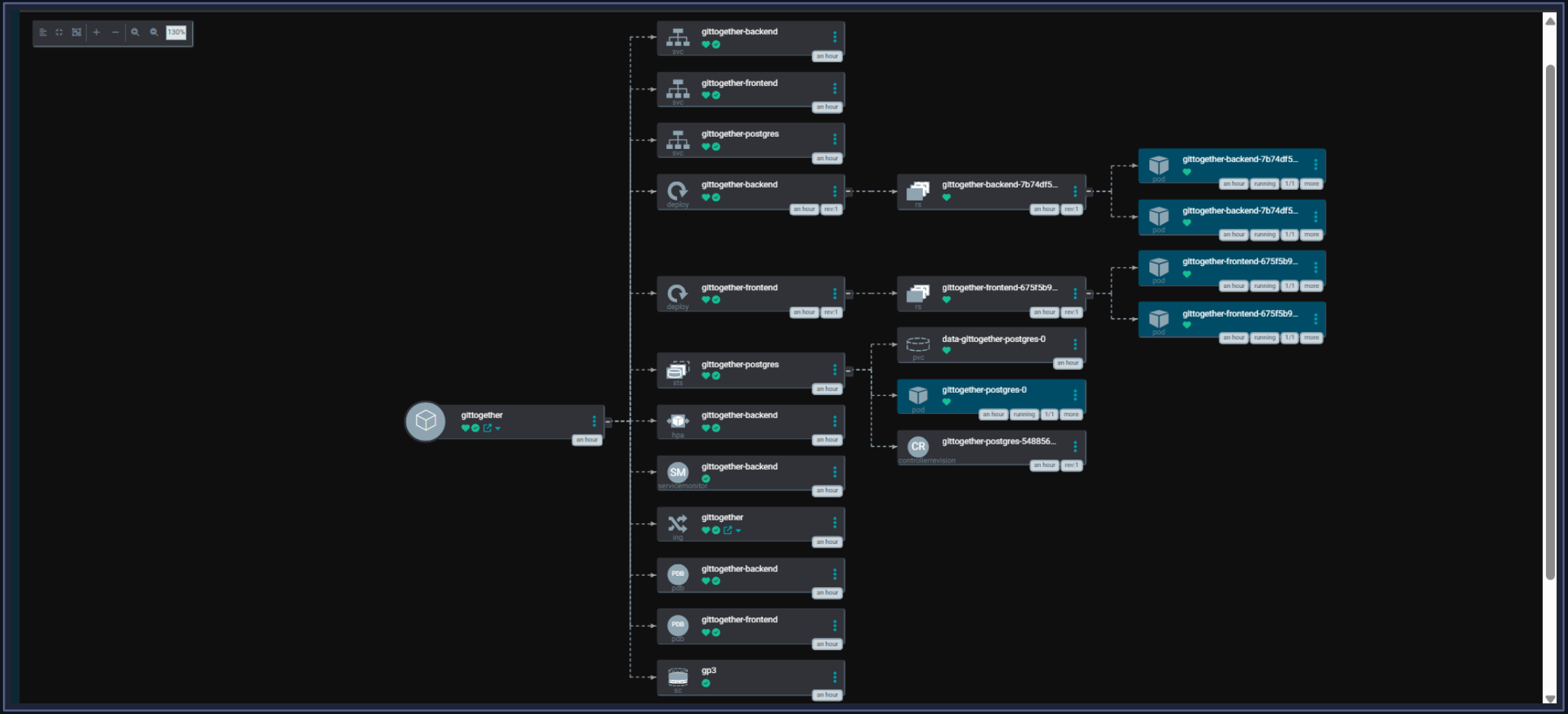
3 new backend pods, created by the HPA while the load test ran.

ArgoCD: Synced and Healthy



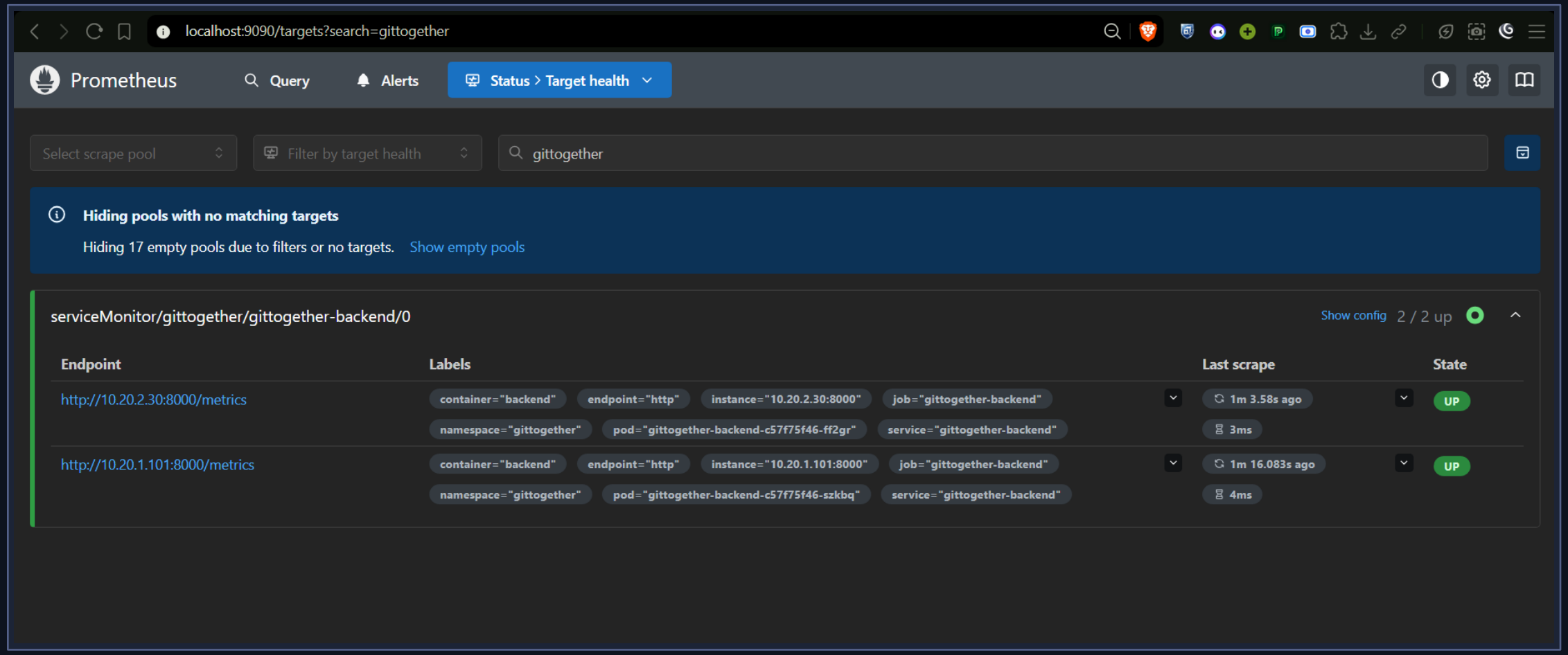
Auto-sync, prune and self-heal from the main branch of the repo.

What ArgoCD manages



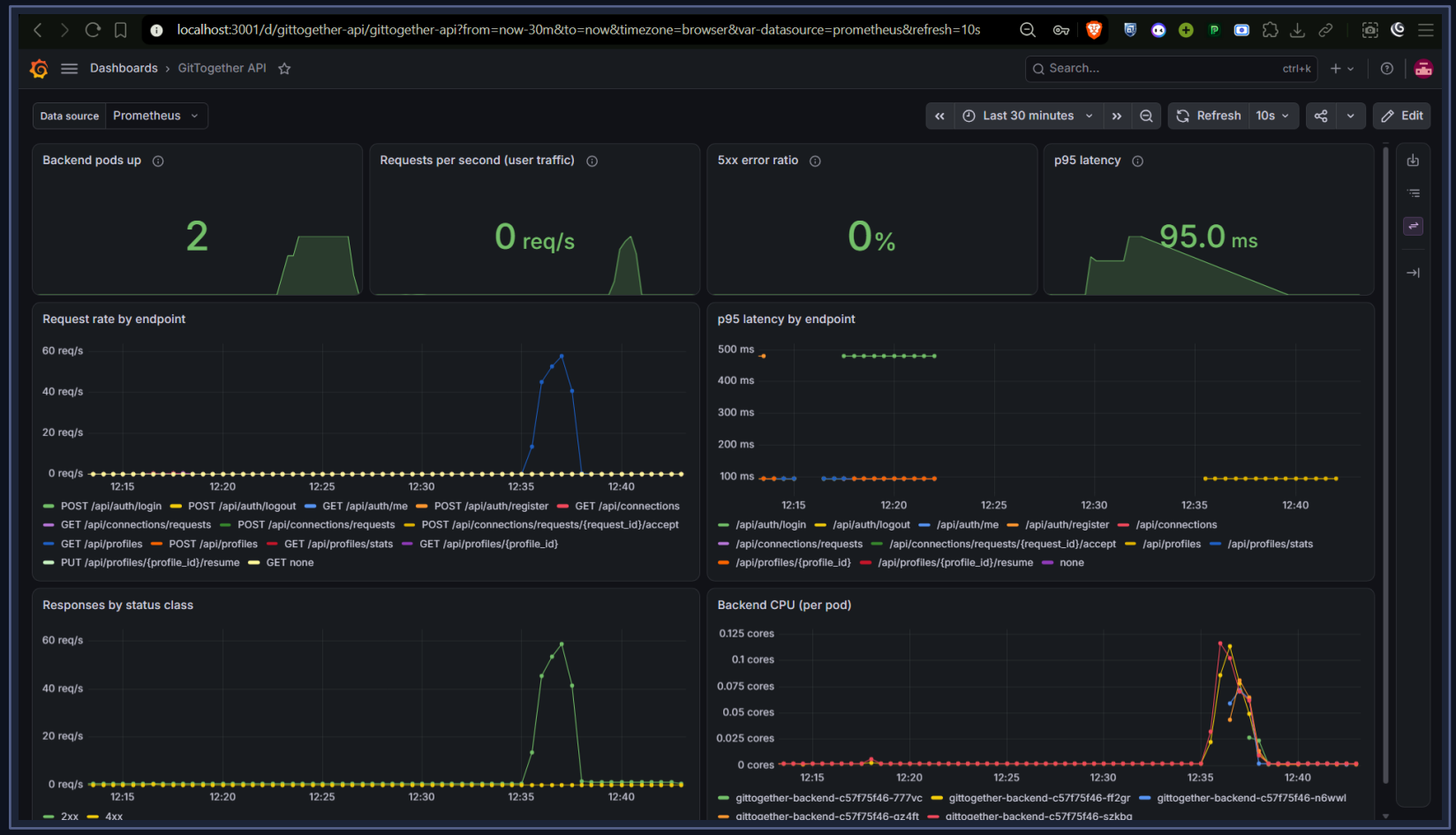
Every resource of the chart, from services to pods to the StorageClass, all green.

Prometheus targets



Both backend pods UP, scraped through a ServiceMonitor.

Grafana dashboard under load



Requests, errors, latency and CPU, provisioned from a ConfigMap in the repo.

CLEANUP

Destroyed the same day

```
module.eks.aws_eks_cluster.this[0]: Still destroying... [id=gittogether-eks, 01m40s elapsed]
module.eks.aws_eks_cluster.this[0]: Still destroying... [id=gittogether-eks, 01m50s elapsed]
module.eks.aws_eks_cluster.this[0]: Still destroying... [id=gittogether-eks, 02m00s elapsed]
module.eks.aws_eks_cluster.this[0]: Still destroying... [id=gittogether-eks, 02m10s elapsed]
module.eks.aws_eks_cluster.this[0]: Still destroying... [id=gittogether-eks, 02m20s elapsed]
module.eks.aws_eks_cluster.this[0]: Still destroying... [id=gittogether-eks, 02m30s elapsed]
module.eks.aws_eks_cluster.this[0]: Still destroying... [id=gittogether-eks, 02m40s elapsed]
module.eks.aws_eks_cluster.this[0]: Destruction complete after 2m42s
module.eks.aws_security_group_rule.node["ingress_self_coredns_udp"]: Destroying... [id=sgrule-4276764746]
module.eks.aws_security_group_rule.node["ingress_cluster_9443_webhook"]: Destroying... [id=sgrule-840390779]
module.eks.aws_security_group_rule.node["ingress_self_coredns_tcp"]: Destroying... [id=sgrule-1521869733]
module.eks.aws_security_group_rule.node["ingress_cluster_443"]: Destroying... [id=sgrule-2307628639]
module.eks.aws_security_group_rule.node["ingress_cluster_10251_webhook"]: Destroying... [id=sgrule-1082911586]
module.eks.aws_security_group_rule.cluster["ingress_nodes_443"]: Destroying... [id=sgrule-2664406179]
module.eks.aws_security_group_rule.node["ingress_cluster_8443_webhook"]: Destroying... [id=sgrule-1454795696]
module.eks.module.kms.aws_kms_key.this[0]: Destroying... [id=dd090e6e-0cbf-4bd2-9579-fa3db5bfd076]
module.eks.aws_security_group_rule.node["egress_all"]: Destroying... [id=sgrule-2780235464]
module.vpc.aws_subnet.private[1]: Destroying... [id=subnet-058f58d2390c0cd1e]
module.eks.module.kms.aws_kms_key.this[0]: Destruction complete after 0s
module.eks.aws_security_group_rule.node["ingress_cluster_4443_webhook"]: Destroying... [id=sgrule-2359832583]
module.vpc.aws_subnet.private[1]: Destruction complete after 1s
module.eks.aws_iam_role_policy_attachment.this["AmazonEKSClusterPolicy"]: Destroying... [id=gittogether-eks-cluster-4ed85ac1f36beac29f5024f5eb/arn:aws:iam::aws:policy/Ama
zonEKSClusterPolicy]
module.eks.aws_security_group_rule.cluster["ingress_nodes_443"]: Destruction complete after 1s
module.vpc.aws_subnet.private[0]: Destroying... [id=subnet-0ec5d1f327bcf59ad]
module.eks.aws_security_group_rule.node["ingress_cluster_443"]: Destruction complete after 1s
module.eks.aws_security_group_rule.node["ingress_nodes_ephemeral"]: Destroying... [id=sgrule-2638403806]
module.vpc.aws_subnet.private[0]: Destruction complete after 0s
module.eks.aws_security_group_rule.node["ingress_cluster_kubelet"]: Destroying... [id=sgrule-1958918937]
module.eks.aws_security_group_rule.node["ingress_self_coredns_udp"]: Destruction complete after 1s
module.eks.aws_security_group_rule.node["ingress_cluster_6443_webhook"]: Destroying... [id=sgrule-1172985745]
module.eks.aws_security_group_rule.node["ingress_cluster_10251_webhook"]: Destruction complete after 2s
module.eks.aws_iam_role_policy_attachment.this["AmazonEKSClusterPolicy"]: Destruction complete after 1s
module.eks.aws_iam_role.this[0]: Destroying... [id=gittogether-eks-cluster-4ed85ac1f36beac29f5024f5eb]
module.eks.aws_security_group_rule.node["ingress_self_coredns_tcp"]: Destruction complete after 2s
module.eks.aws_security_group_rule.node["ingress_cluster_9443_webhook"]: Destruction complete after 3s
module.eks.aws_iam_role.this[0]: Destruction complete after 1s
module.eks.aws_security_group_rule.node["egress_all"]: Destruction complete after 3s
module.eks.aws_security_group_rule.node["ingress_cluster_8443_webhook"]: Destruction complete after 4s
module.eks.aws_security_group_rule.node["ingress_cluster_4443_webhook"]: Destruction complete after 4s
module.eks.aws_security_group_rule.node["ingress_nodes_ephemeral"]: Destruction complete after 4s
module.eks.aws_security_group_rule.node["ingress_cluster_kubelet"]: Destruction complete after 4s
module.eks.aws_security_group_rule.node["ingress_cluster_6443_webhook"]: Destruction complete after 4s
module.eks.aws_security_group.cluster[0]: Destroying... [id=sg-0257535de5074e888]
module.eks.aws_security_group.node[0]: Destroying... [id=sg-06f2e5f477d945860]
module.eks.aws_security_group.node[0]: Destruction complete after 1s
module.eks.aws_security_group.cluster[0]: Destruction complete after 1s
module.vpc.aws_vpc.this[0]: Destroying... [id=vpc-04cd1d9c346089dbf]
module.vpc.aws_vpc.this[0]: Destruction complete after 1s

Destroy complete! Resources: 59 destroyed.
```

59 destroyed. The cluster costs about $0.40 an hour, so it only exists while I use it.

LESSONS

What went wrong, and what I learned

1

Free plan limits

t3.medium was rejected and the account allows only 5 vCPUs. Fix: 2 x m7i-flex.large.

Check quotas before designing.

2

EBS driver crash loop

Pods could not use the node's credentials, so Postgres had no disk. Fix: EKS Pod Identity.

Read the logs, don't guess.

3

ArgoCD on EKS

The API server returned 503 for the schema. Fix: unregister the two metrics APIs while ArgoCD starts.

Reproduce on a clean cluster.

NEXT STEPS

What I would do next

1. Managed database (Amazon RDS) with backups
2. HTTPS with a real domain and secure cookies
3. CI step that bumps the image tag automatically
4. Restrict the EKS API endpoint; add rate limiting

github.com/AhmadAnzar/GitTogether